

Semester: **Fall 2023**
Course Code: **CSE460**
Course Title: **VLSI Design**

Final Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 30 minutes**
Date: **15th Dec 2023**

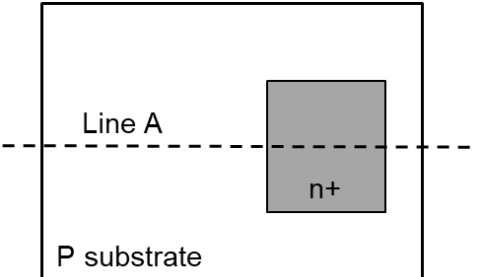
Set A

Student ID:	Name:	Section:
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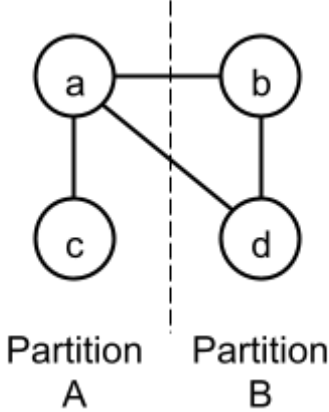
[Answer **both** questions **1** and **2**. You can answer any **one** question from **questions 3** and **4**. In total, you need to answer **three** questions out of **four**.]

[Each question carries equal marks.]

[After the exam, *the question paper should be turned in along with the answer script.*]

1. (CO3)	(a)	Consider a CMOS compound gate that implements the following function: $Y = \overline{A + BC + D}$	
		(i) Draw the stick diagram of the given CMOS gate.	7
		(ii) Find the total area of the stick diagram in (i) in terms of Lambda (λ).	3
	(b)	Observe Figure 1 and answer the following questions.	
		 Figure 1: Top view of a wafer after fabricating an n⁺ diffusion region (dark region)	
		(i) Briefly explain why n-well is required in CMOS fabrication.	3
		(ii) Draw the cross-sectional view of Figure 1 along 'Line A' and label the different regions.	2

2. (CO1)	Consider the following truth tables for two logic gates																																
	<table><tr><td>A</td><td>B</td><td>Y</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Y	0	0	1	0	1	1	1	0	1	1	1	0	<table><tr><td>A</td><td>B</td><td>Y</td></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Y	0	0	1	0	1	0	1	0	0	1	1	0	
	A	B	Y																														
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Gate 1		Gate 2																															
(a)	Identify the logic gates and draw their CMOS circuit representation.		2																														
(b)	Determine the individual transistor widths k_{nMOS} and k_{pMOS} for both of the gates to achieve the effective rise and fall resistances equal to that of a unit inverter, R , in the worst case.		2																														
(c)	Draw the simplified RC circuit only for Gate 1 and determine the input capacitance of Gate 2 in terms of C		4+1																														
(d)	Consider Gate 1 is driving 12 identical Gate 2 circuits (only one of the inputs of each). Now derive the expressions for t_{pdf} and t_{cdf} by sketching the corresponding RC networks from (c).		6																														
3. (CO4)	For the 4×4 maze below, the dark regions are obstacles or components.																																
	<table><tr><td></td><td></td><td></td><td>T₁</td></tr><tr><td>S</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td>T₂</td></tr></table>						T₁	S											T₂														
				T₁																													
S																																	
			T₂																														
The total memory requirements for the maze is 4 bytes .																																	
(a)	Calculate the number of bits per cell (n) and the maximum value that a cell can store (L) as per the memory requirements		2+1																														

	(b)	State the sequence for wave propagation as per the memory requirements. Perform wave propagation to find the shortest path from source, S to both of the targets, T₁ and T₂	2+6
	(c)	Show the shortest path for each of the targets.	4
4. (CO4)	The graph below (nodes a-d) can be optimally partitioned using the Kernighan-Lin algorithm. The dotted line represents the initial partitioning. Assume all the edges have the same weight.  Partition A Partition B		
	(a)	Calculate the initial cut cost.	2
	(b)	Identify how many iterations are needed in a single pass?	2
	(c)	Perform the first pass of the algorithm and determine the new cut cost of the optimum output of the first pass. [<u>Hint</u>: For the “i”th iteration of the first pass, until all the nodes are swapped and fixed, do the following: • Compute the node costs of all unfixed nodes • Find the maximum gain of swapping a pair of nodes (Δg_i) • Swap the pair and draw the updated graph • Calculate cumulative gain, G_i]	9
	(d)	Should you perform subsequent passes of the algorithm? Why or why not?	2